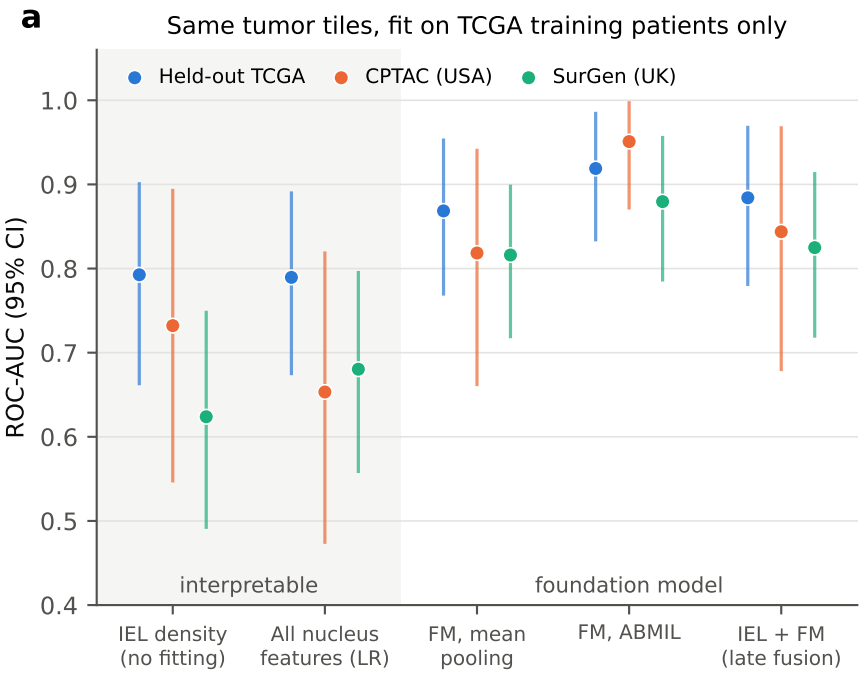
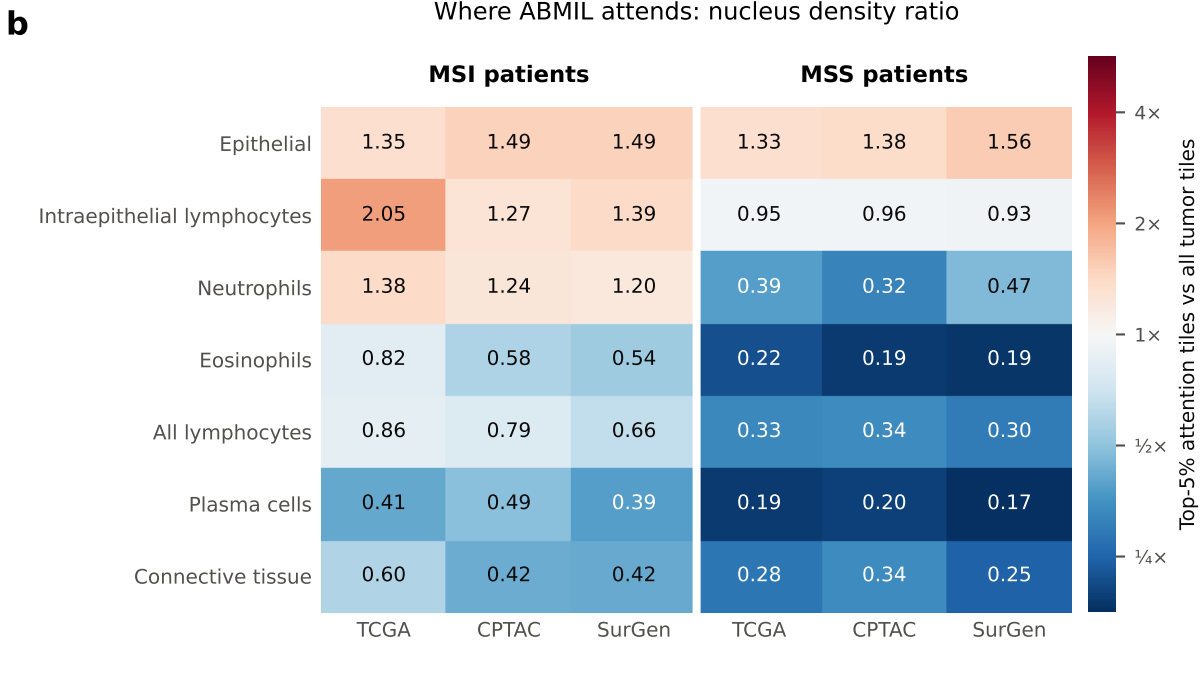

a
Same tumor tiles, fit on TCGA training patients only
Held-out TCGA
CPTAC (USA)
SurGen (UK)
ROC-AUC (95% CI)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
interpretable
foundation model
IEL density (no fitting)
All nucleus features (LR)
FM, mean pooling
FM, ABMIL
IEL + FM (late fusion)


b
Where ABMIL attends: nucleus density ratio
MSI patients
MSS patients
Epithelial 1.35 1.49 1.49 1.33 1.38 1.56
Intraepithelial lymphocytes 2.05 1.27 1.39 0.95 0.96 0.93
Neutrophils 1.38 1.24 1.20 0.39 0.32 0.47
Eosinophils 0.82 0.58 0.54 0.22 0.19 0.19
All lymphocytes 0.86 0.79 0.66 0.33 0.34 0.30
Plasma cells 0.41 0.49 0.39 0.19 0.20 0.17
Connective tissue 0.60 0.42 0.42 0.28 0.34 0.25
TCGA
CPTAC
SurGen
TCGA
CPTAC
SurGen
Top-5% attention tiles vs all tumor tiles
4×
2×
1×
½×
¼×